Recommended Assessment
Servomotor Lab

Pulse Width Modulation (PWM)

1. When the duty cycle was set to 50% in the visualization tool, what pixel value do you expect for the gray color in Part B of the visualization?

2. As you increase or decrease the duty cycle, what pixel values did you see for the gray color in Part B of the visualization? Why?

3. Why does the section in part A flicker and differ from the average color/brightness in part B?

Driving a Servomotor

4. What was the range specification for the servomotor you used from the datasheet? Based on this, write a simple equation that represents the conversion from your user command $\boldsymbol{p}$ (0 to 1) to the corresponding servo range $\boldsymbol{r}$ used in the firmware.

5. Modify the equation from the previous example to output the servo angle $\boldsymbol{\theta}$ instead of the duty cycle $\boldsymbol{r}$ based on the range of motion of your servomotor and the user command $\boldsymbol{p}$.

6. What bias did you observe for your servomotor? Update the equation converting the user command $\boldsymbol{p}$ to a servo angle $\boldsymbol{\theta}$ incorporating the bias $\boldsymbol{b}$.